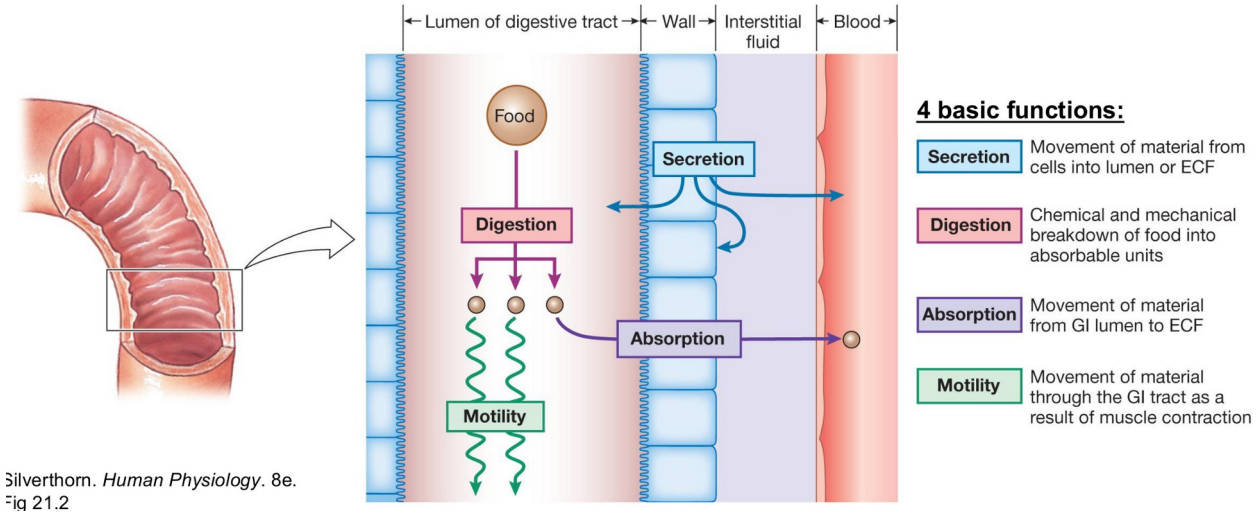
01 / Digestive physiology: the working map

Lecture: BBMS2006_2026-27_Digestive Physiology.pdf, 28 PDF pages | Tutorial: GI Physiology Tutorial 2026-27.pdf, 5 PDF pages

Follow the meal

Food cues → neural preparation → gastric mixing and acid secretion → duodenal feedback → pancreatic secretion → digestion → absorption.



Original: four basic GI functions (lecture p. 3; Silverthorn, Human Physiology, 8e, Fig. 21.2).

Function	Meaning
Secretion	Move water, ions, acid, mucus or enzymes into the GI lumen.
Digestion	Break large nutrients into smaller absorbable products.
Absorption	Move substances from the lumen across epithelium into the internal environment.
Motility	Mix luminal contents and move them through the tract.

The lumen is continuous with the external environment. A nutrient is absorbed only after it crosses the epithelium. GI = gastrointestinal; ECF = extracellular fluid.

Where the supplied questions fit

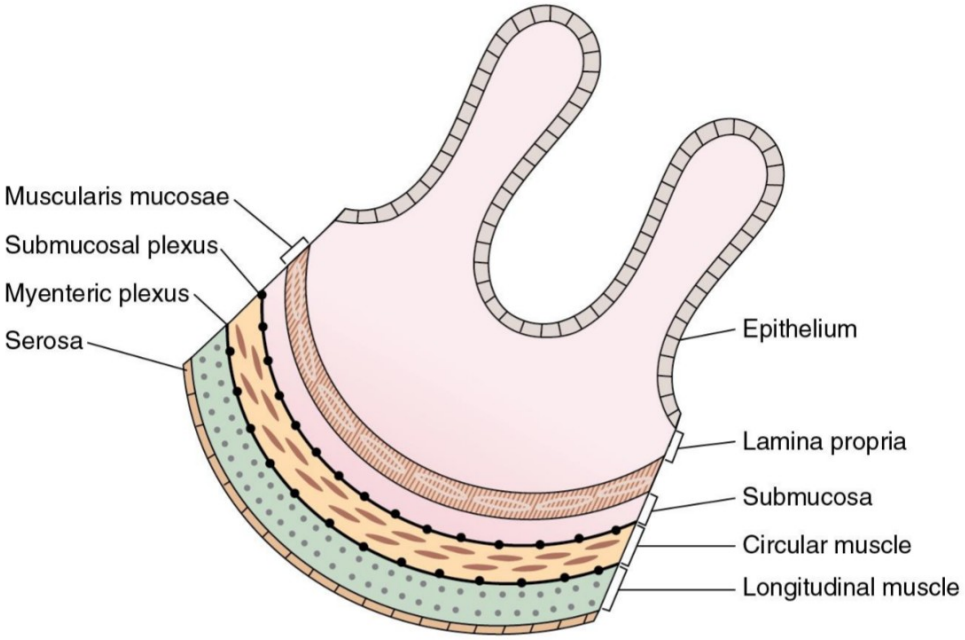
Notes pages	Original questions integrated there
2–4 | Wall, ENS and motility	MCQ 1–2; foundations for both cases.
5–9 | Gastric secretion	Case 1, all six questions; Case 2 Parts 1–2; MCQ 3.
10–11 | Duodenum and pancreas	MCQ 4–6; Case 2 Part 3.
12–16 | Digestion and absorption	MCQ 7; Case 2 Part 4; two integrated practice questions occur on pp. 9 and 16.

Case context: Case 1 follows a student watching food videos before eating. Case 2 follows a mixed buffet meal: rice, steak, fried chicken, chips, ice cream and soft drinks.

All seven MCQs and both cases are relevant. Case 2 is headed “Part D” in the source; its four internal parts are retained here. Answers are independently reasoned suggestions, not an official answer key. [Extension] marks useful detail beyond the slide text. Original teaching figures are retained; duplicate figures appear once. Decorative logos are omitted. PDF page numbers refer to the supplied files.

02 / GI wall and muscle actions

Lecture PDF pp. 4 and 7 | Same wall figure appears on both source slides



Original: GI wall, muscle layers and nerve plexuses (pp. 4/7; Costanzo Physiology, 7e, Fig. 8.1).

From lumen outward	Main role / location
Mucosa: epithelium	Barrier; secretion and absorption.
Mucosa: lamina propria	Supportive connective tissue beneath the epithelium.
Mucosa: muscularis mucosae	Thin local smooth-muscle layer; distinct from muscularis externa.
Submucosa	Support, vessels and submucosal nerve plexus.
Muscularis externa	Inner circular and outer longitudinal muscle; major mixing and propulsion.
Myenteric plexus	Between circular and longitudinal muscle; primarily regulates motility.

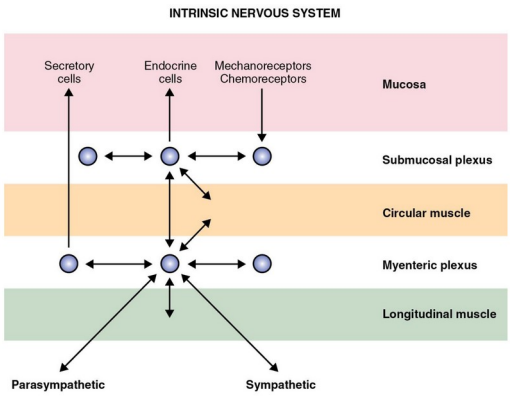
Contraction: direction and time course

Term	What changes?	Useful example
Circular contraction	Lumen diameter decreases.	Local narrowing splits / mixes contents.
Longitudinal contraction	Segment length decreases.	Shortening coordinates with circular muscle.
Phasic contraction	Contraction alternates with relaxation.	Repeated mixing contractions.
Tonic contraction	Sustained baseline tension.	Sphincter tone controls passage.

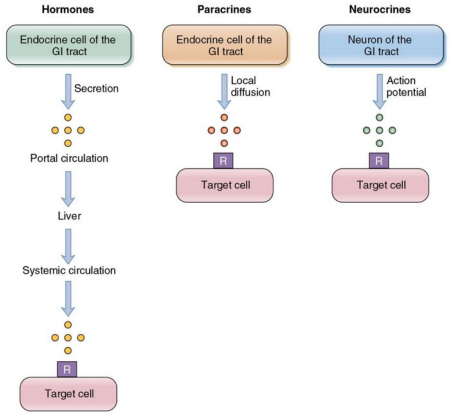
Do not mix up the two muscle layers
Muscularis mucosae is a thin part of the mucosa. Muscularis externa contains the circular and longitudinal layers shown around the myenteric plexus.

03 / ENS, autonomic input and messengers

Lecture PDF pp. 5–6 | Original MCQs 1–2



Original ENS pathways (p. 5; Costanzo Fig. 8.3).



Original signalling routes (p. 6; Costanzo Fig. 8.4).

Control	Key effect
ENS = enteric nervous system	Local circuits can function independently; autonomic input modulates them.
Myenteric plexus	Mainly controls motility. Submucosal plexus mainly controls secretion and local mucosal function.
Parasympathetic input	Generally increases GI motility and secretion.
Sympathetic input	Generally decreases GI motility and secretion.

Route	How it reaches its target	Example
Neurocrine	Neuron → nearby target cell.	Acetylcholine (ACh).
Endocrine	Enteroendocrine cell → blood → target.	Gastrin, CCK, secretin.
Paracrine	Local release → neighbouring cells.	Histamine, somatostatin in the stomach.

ACh and histamine are not peptides, despite the slide heading “GI peptides”. Signalling route and chemical class are different classifications.

Original MCQ 1

Which component of the enteric nervous system primarily regulates GI motility?

A. Submucosal plexus B. Myenteric plexus C. Parietal cells D. ECL cells

Answer: B

Myenteric = motility. Submucosal = mainly secretion. Parietal and ECL cells are gastric secretory cells, not nerve plexuses.

Original MCQ 2

Which of the following generally increases GI motility and secretion?

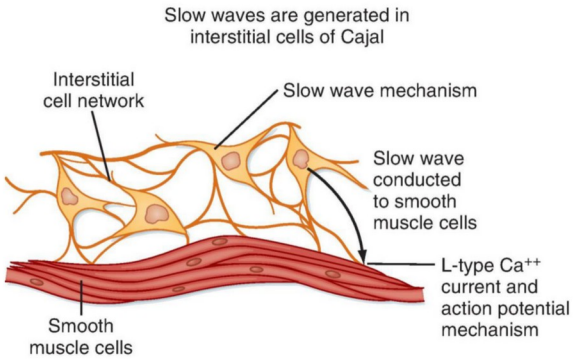
A. Sympathetic stimulation B. Somatostatin C. Parasympathetic stimulation D. Secretin

Answer: C

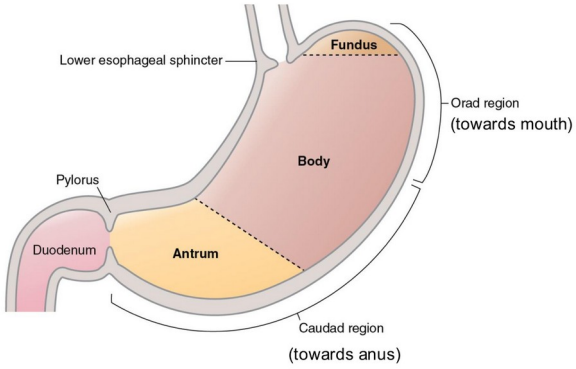
Parasympathetic input generally promotes both. Individual muscle and sphincter responses can differ.

04 / Pacemakers, gastric movement and mixing

Lecture PDF pp. 8–10



Original ICC–smooth-muscle coupling (p. 8; Berne & Levy, 8e, Fig. 29.12).



Costanzo Physiology, 7e, Fig 8.10

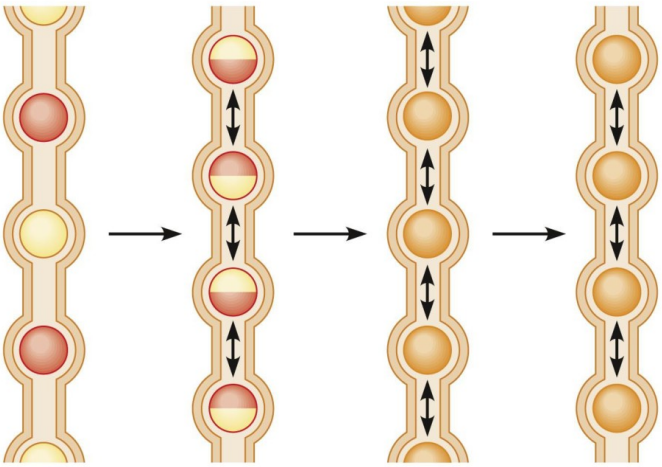
Original stomach regions (p. 9; Costanzo Fig. 8.10).

Electrical timing → muscle activity

Interstitial cells of Cajal (ICCs) generate slow-wave electrical activity.
Gap junctions spread activity to smooth muscle → coordinated contraction.
ICCs are pacemaker cells, distinct from enteric neurons.

[Extension] Slow waves set the timing of excitability. Not every slow wave produces a strong contraction; sufficient depolarisation permits Ca^{2+} spikes and contraction. Autonomic and enteric signals modify excitability and force.

Gastric task	Mechanism / result
1. Receive and store	Orad (toward the mouth; proximal stomach) relaxes to accommodate food.
2. Grind and mix	Contractions reduce particle size and mix food with gastric secretions.
3. Empty	Coordinated contractions deliver chyme through the pylorus into the small intestine.



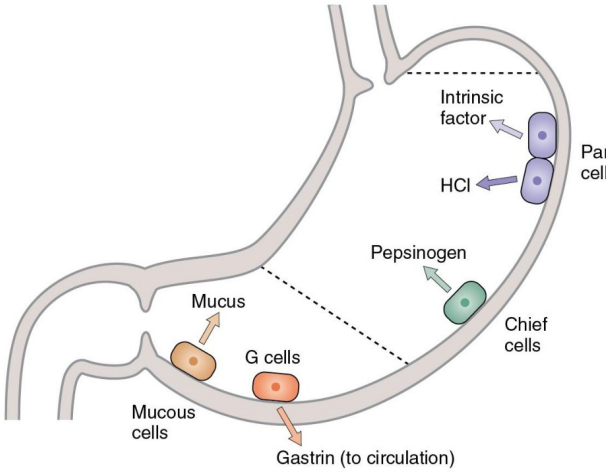
Original segmentation sequence (p. 10; Stanfield, Fig. 20.30b).

Segmentation: alternating circular contractions → split chyme → move it back and forth → rejoin and mix. It improves contact with enzymes and the absorptive surface.

Main role = mixing, rather than sustained net propulsion. The slide's "does not propel" is a teaching simplification. Caudad means toward the anus; chyme is the partly digested mixture leaving the stomach.

05 / Gastric juice: match the cell to its product

Lecture PDF p. 11 | Case 2 Part 2, Q3–4



Original gastric glands (p. 11; Costanzo Fig. 8.14).

Cell Type	Location	Secretion
Parietal cells	Body	HCl Intrinsic factor (to lumen)
Chief cells	Body	Pepsinogen (to lumen)
G cells	Antrum	Gastrin (circulation)
Mucous cells	Antrum	Mucus (to lumen)

Original cell–product table with lumen / blood annotations (p. 11).

Cell / product	Destination and function
Parietal cells → HCl	Into the lumen. Acid denatures proteins and supports activation / action of pepsin.
Parietal cells → intrinsic factor	Into the lumen. Required for vitamin B_{12} absorption in the ileum.
Chief cells → pepsinogen	Into the lumen. Inactive precursor converted to pepsin in acid; pepsin begins protein digestion.
Mucous cells → mucus	Into the lumen / mucus layer. Protection and lubrication; the mucosal barrier limits acid injury.
Antral G cells → gastrin	Into blood. Endocrine stimulation of gastric acid secretion; not a luminal digestive enzyme.

Original Case 2 | Part 2: Gastric Acid Secretion

Original question / ID	Answer and reasoning
C2-P2-Q3 Which gastric cells secrete hydrochloric acid?	Parietal cells. Do not substitute chief cells, which secrete pepsinogen.
C2-P2-Q4 State ONE physiological role of gastric acid.	One sufficient answer: promotes conversion of pepsinogen to pepsin. Alternatives: maintains acidic conditions for pepsin, denatures proteins, or helps kill ingested microbes.

Useful separation
HCl is acid; pepsin is the proteolytic enzyme; pepsinogen is its inactive precursor.
Intrinsic factor assists later vitamin B_{12} absorption; it does not digest proteins.

06 / Cephalic and gastric phases: start signals

Lecture PDF p. 12, checked against p. 15 | Original Case 1; Case 2 Part 1

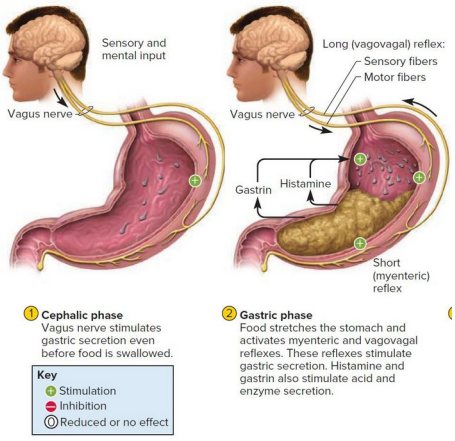


FIGURE 25.17 Neural and Hormonal Control of the Stomach.

Original cephalic / gastric reflex diagram (p. 12; Saladin, Anatomy & Physiology, 10e).

Cephalic: before food arrives

Sight / smell / taste / anticipation → CNS (central nervous system) → vagal parasympathetic output → preparation of gastric secretion and motility.

Gastric: food in stomach

Distension → mechanoreceptors → local ENS and long vagovagal reflexes → ACh. Peptides / amino acids promote gastrin; meal buffering initially reduces acid inhibition.

Correction: low gastric pH inhibits

Slide 12 and the handwritten buffet answer are misleading. Strong gastric acidity → D-cell somatostatin → acid secretion falls. Distension and food peptides / amino acids stimulate the gastric phase.

2. What stimulus initiates this phase?

low pH

Original Case 2 annotation, retained for comparison; "low pH" is corrected above (tutorial p. 4).

Original Case 1 | Food Influencer Effect

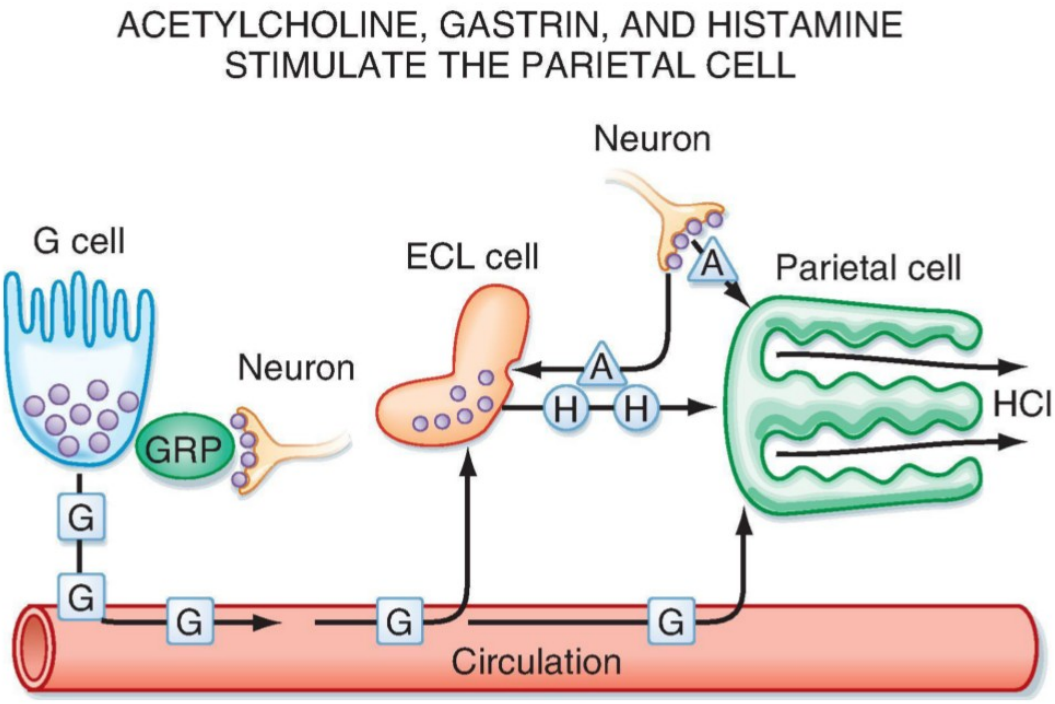
Original question / ID	Answer and reasoning
C1-Q1 Which phase of gastric secretion is activated?	Cephalic phase: food cues are present, but no meal has entered the stomach.
C1-Q2 Which autonomic branch is primarily responsible?	Parasympathetic, chiefly via the vagus nerve.
C1-Q3 Which neurotransmitter is released?	ACh is the expected general answer. GRP specifically stimulates G-cell gastrin release (p. 7).
C1-Q4 Effect on gastric motility?	Generally increases preparatory gastric motility.
C1-Q5 Effect on gastric acid secretion?	Increases gastric acid secretion.
C1-Q6 One advantage of activation before food enters?	Feedforward control prepares secretions and motility, reducing the delay once food arrives.

Original Case 2 | Part 1: Gastric Phase

Original question / ID	Answer and reasoning
C2-P1-Q1 Which phase becomes dominant after the meal?	Gastric phase.
C2-P1-Q2 What stimulus initiates this phase?	Gastric distension; food peptides / amino acids also stimulate secretion. Do not answer simply "low pH".
C2-P1-Q3 Which neurotransmitter promotes motility and secretion?	ACh.

07 / Three stimulators converge on acid secretion

Lecture PDF pp. 13–14 | Case 2 Part 2, Q1



Original neural, gastrin and histamine pathways (same figure on pp. 13–14; Berne & Levy, 8e, Fig. 29.7).

Messenger / origin	Route and target	Effect
ACh: enteric neurons	Neurocrine → parietal M_3 receptors; also stimulates ECL-cell activity.	Promotes H^+ secretion directly and through histamine.
Gastrin: antral G cells	Endocrine → blood → CCK_2 receptors on ECL / parietal cells. CCK_2 is also called CCK-B.	ECL histamine release ↑; parietal acid secretion ↑.
Histamine: ECL cells	Paracrine → nearby parietal H_2 receptors.	Proton-pump activity and acid secretion ↑.

Neural control has two relevant transmitters

Vagal pathways recruit enteric neurons.
ACh → parietal / ECL stimulation.
Gastrin-releasing peptide (GRP) → G cells → gastrin in blood.
Do not replace GRP with ACh when asked specifically how vagal input stimulates G-cell gastrin release.

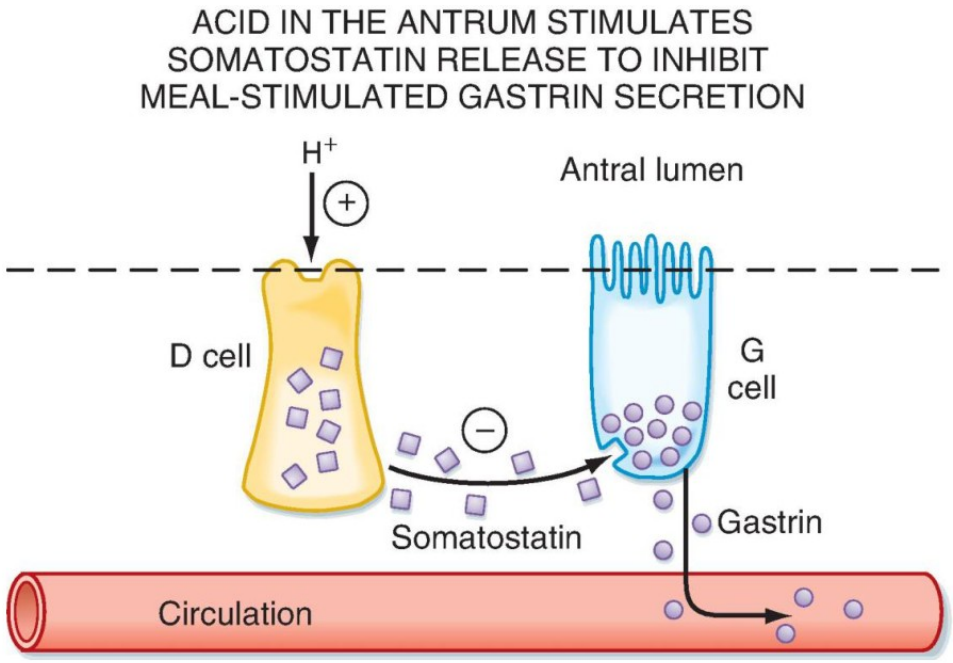
Original Case 2 | Part 2, Q1

Original question / ID	Answer and reasoning
C2-P2-Q1 Name THREE stimulators of gastric acid secretion.	ACh + gastrin + histamine. The source annotation lists gastrin and histamine; ACh completes the required three.

ECL = enterochromaffin-like cell. Gastrin and CCK are distinct hormones even though their receptor names share "CCK". H^+/K^+-ATPase is the parietal proton pump; it uses ATP to secrete H^+ in exchange for K^+.

08 / Somatostatin: the acid feedback brake

Lecture PDF p. 15 | Original MCQ 3; Case 2 Part 2, Q2



Original D-cell inhibitory pathways (p. 15; Berne & Levy, 8e, Fig. 29.9).

Follow the feedback loop

Food buffers secreted H^+ → the fall in gastric pH is initially limited.
Food leaves → buffering decreases → gastric pH falls more.
Low gastric pH → D cells → somatostatin → acid secretion decreases.

Somatostatin target	Result
Parietal cells	Direct inhibition of HCl secretion.
ECL cells	Histamine release ↓ → less stimulation of parietal cells.
G cells	Gastrin release ↓ → less stimulation of ECL and parietal cells.

Original MCQ 3

Which regulator directly inhibits gastric acid secretion?

A. Gastrin B. Histamine C. Acetylcholine D. Somatostatin

Answer: D

Somatostatin inhibits parietal cells directly. It also inhibits gastrin and histamine release. The other three options stimulate acid secretion.

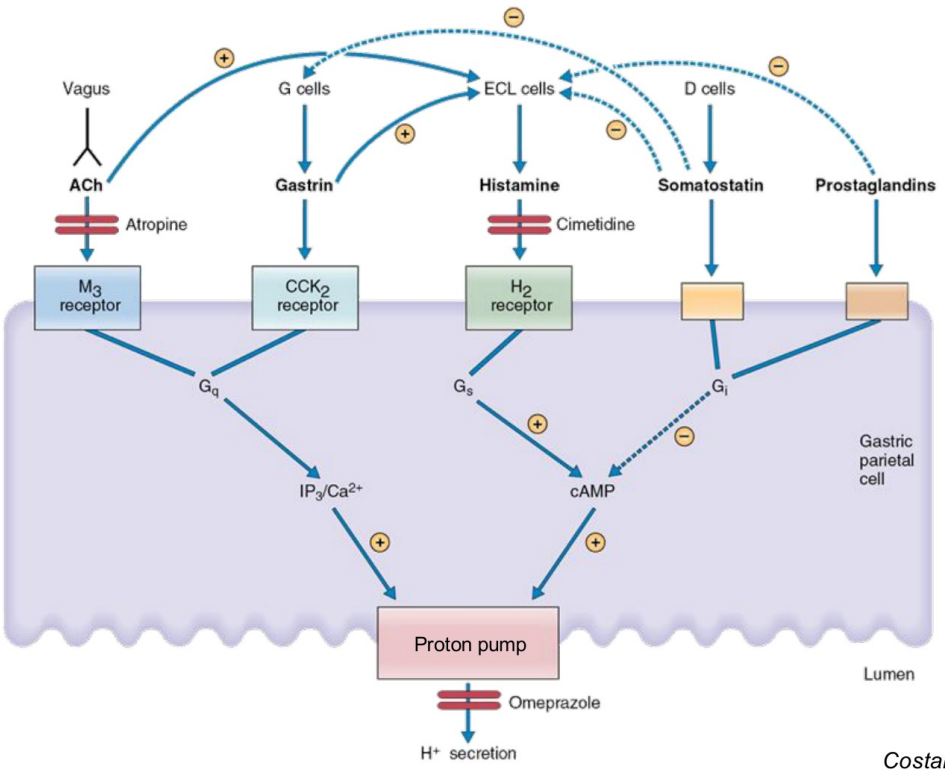
Original Case 2 | Part 2, Q2

Original question / ID	Answer and reasoning
C2-P2-Q2 Name ONE inhibitor of gastric acid secretion.	Somatostatin. Link it to gastric D cells and the low-pH negative-feedback loop.

"Direct" asks whether the mediator can act on the parietal cell itself. It does not mean the mediator has no additional indirect effects.

09 / Acid receptors: integrate inputs and feedback

Lecture PDF p. 16 | One integrated practice question based on the original diagram



Costanzo Physiology. 7e. Fig 8.17

Original receptor and drug-target summary (p. 16; Costanzo Physiology, 7e, Fig. 8.17).

Stimulatory route	Intracellular signal → final step
ACh → M_3; gastrin → CCK_2	Gq → IP_3 / Ca^{2+} → promote proton-pump activity.
Histamine → H_2	Gs → cAMP → promote proton-pump activity.
Somatostatin / prostaglandins	Inhibitory Gi signalling lowers cAMP and acid secretion.

cAMP = cyclic adenosine monophosphate; IP_3 = inositol trisphosphate. Gq / Gs / Gi are intracellular G-protein signalling classes. The diagram shows atropine at muscarinic receptors, cimetidine at H_2 receptors and omeprazole at the proton pump.

Integrated practice 1 | Self-written; not an original exam question

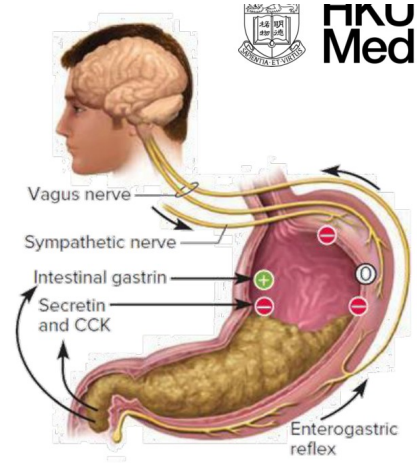
Two treatments block either H_2 receptors or H^+/K^+-ATPase. Can ACh and gastrin still activate their receptors? Why can gastrin rise after sustained acid suppression?

Answer approach:

- H_2 blockade removes histamine signalling. ACh–M_3 and gastrin–CCK_2 inputs remain; loss of histamine also weakens their combined effect.
- Pump inhibition blocks the final acid-secretory step despite upstream receptor signals.
- [Extension] Acid suppression → gastric pH ↑ → D-cell somatostatin feedback ↓ → G-cell inhibition ↓ → gastrin can rise. High gastrin does not guarantee high acid when the pump is inhibited.

10 / Duodenal feedback slows gastric delivery

Lecture PDF p. 17 | Original MCQ 6; Case 2 Part 3, Q1–2 and Q5



3 Intestinal phase
Intestinal gastrin briefly stimulates the stomach, but then secretin, CCK, and the enterogastric reflex inhibit gastric secretion and motility while the duodenum processes the chyme already in it. Sympathetic nerve fibers suppress gastric activity, while vagal (parasympathetic) stimulation of the stomach is now inhibited.

Original intestinal inhibitory phase (p. 17; Saladin, 10e).

Neural brake: enterogastric reflex

Duodenal lipids, acid or distension → gastric motor activity inhibited + pyloric resistance increased → gastric emptying slows.

Nutrient signal: CCK

Fatty acids, peptides and amino acids → I cells in duodenum / jejunum → cholecystokinin (CCK). Emptying slows; pancreatic enzyme secretion increases.

Acid signal: secretin

Low duodenal pH → S cells → secretin. Gastric delivery slows; pancreatic bicarbonate-rich secretion increases.

Original MCQ 6

Which of the following slows gastric emptying?

A. Parasympathetic stimulation B. Acetylcholine C. CCK D. Gastric distension

Answer: C

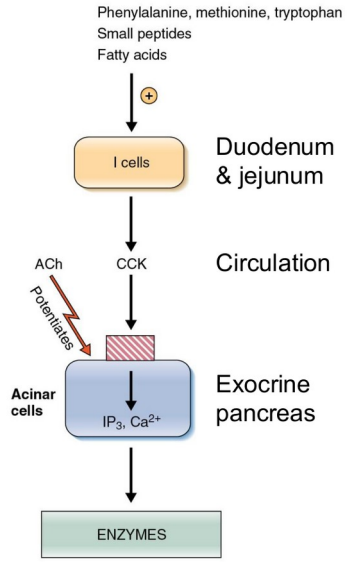
CCK provides intestinal feedback that slows delivery from the stomach. Distension of the stomach and distension of the duodenum are different signals.

Original Case 2 | Part 3: Fatty Chyme Enters the Duodenum

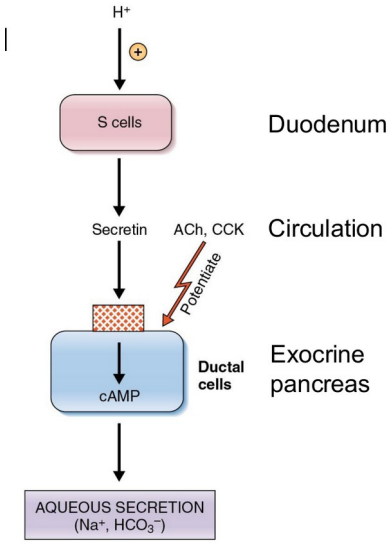
Original question / ID	Answer and reasoning
C2-P3-Q1 Which TWO hormones increase?	CCK in response to lipid / protein products; secretin in response to acid.
C2-P3-Q2 What effect on gastric emptying?	Both slow gastric emptying in this teaching model.
C2-P3-Q5 Why is slowing gastric emptying beneficial?	Limits acid / nutrient overload and allows time for neutralisation, digestion and absorption.

11 / Pancreas: enzymes versus alkaline fluid

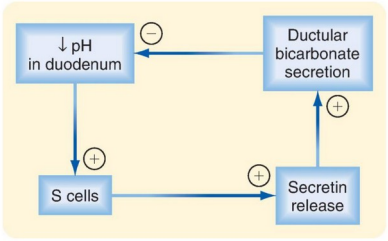
Lecture PDF pp. 18–19 | Original MCQs 4–5; Case 2 Part 3, Q3–4



Original CCK pathway (p. 18; Costanzo Fig. 8.23).



Original secretin pathway (p. 19; Costanzo Fig. 8.23).



Original secretin negative feedback (p. 19; Berne & Levy, 8e, Fig. 30.3).

Compare	CCK	Secretin
Primary stimulus / source	Fatty acids, amino acids, peptides → I cells.	Low duodenal pH → S cells.
Main pancreatic output	Enzyme-rich secretion from acinar cells; CCK_1-mediated pathways.	HCO_3^--rich aqueous secretion from ductal cells.
Purpose	Hydrolyse nutrients into absorbable products.	Neutralise acid and provide a suitable environment for digestive enzymes.
Cooperation	ACh also stimulates enzyme secretion.	ACh and CCK potentiate aqueous secretion; the pathways are not exclusive.

Original MCQs 4–5 | Apply the comparison

Original question	Options	Answer and case link
MCQ 4: Which regulator primarily stimulates pancreatic enzyme secretion?	A. Gastrin B. Secretin C. CCK D. Histamine	C — CCK → enzyme secretion. Also answers C2-P3-Q3: Which hormone stimulates pancreatic enzyme secretion?
MCQ 5: Which regulator primarily stimulates bicarbonate-rich aqueous secretion by the pancreas?	A. Secretin B. CCK C. Acetylcholine D. Gastrin	A — Secretin → bicarbonate-rich fluid. Also answers C2-P3-Q4: Which hormone stimulates bicarbonate-rich aqueous secretion?

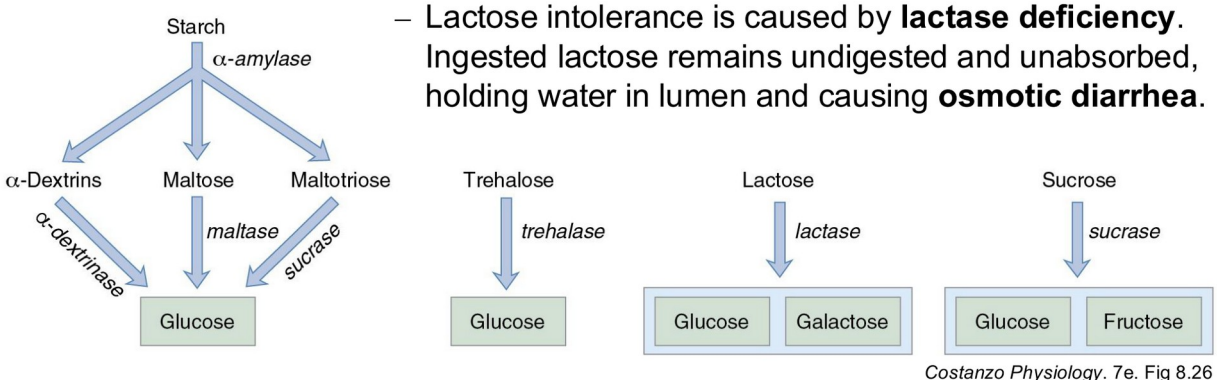
12 / Digestion: enzymes and carbohydrate products

Lecture PDF pp. 20–21 | Corrected enzyme-source table

Nutrient	Salivary / oral and gastric	Pancreas	Intestinal brush border
Carbohydrate	Salivary α-amylase begins starch digestion; no major gastric carbohydrase.	α-Amylase acts in the lumen.	Sucrase, maltase, lactase, trehalase; α-dextrinase / isomaltase.
Protein	Gastric pepsin begins digestion.	Trypsin, chymotrypsin, carboxypeptidases.	Amino-oligopeptidases and dipeptidases; enteropeptidase activates trypsinogen.
Lipid	Lingual lipase provides an early contribution; pancreatic digestion dominates here.	Lipase + colipase; phospholipase A_2; cholesterol ester hydrolase.	No major brush-border lipid hydrolase in this lecture's pathway.

Correction: intestinal α-amylase versus brush-border enzymes

Slides 20–21 attribute a major α-amylase role to intestinal mucosa. Use this distinction: salivary / pancreatic α-amylase → luminal starch digestion; brush-border α-dextrinase / isomaltase and disaccharidases → final monosaccharides. The original pathway figure below correctly labels α-dextrinase.



Original starch and disaccharide pathways, including the lactase-deficiency note (p. 21; Costanzo Physiology, 7e, Fig. 8.26).

Substrate	Enzyme → absorbable products
Starch	α-Amylase → maltose, maltotriose, α-limit dextrins → brush-border enzymes → glucose.
Lactose	Lactase → glucose + galactose.
Sucrose	Sucrase → glucose + fructose.
Trehalose	Trehalase → two glucose molecules.

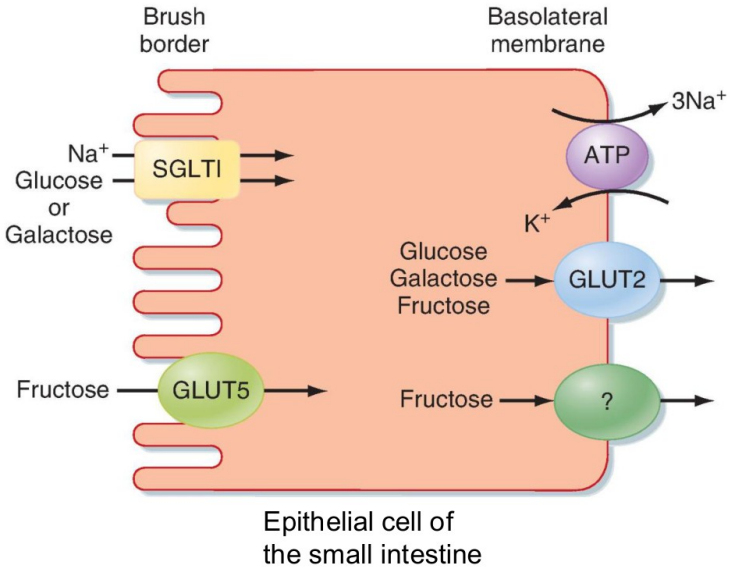
Lactase deficiency → osmotic diarrhoea

Unhydrolysed lactose stays in the lumen → is not absorbed as a disaccharide → retains water → osmotic diarrhoea. [Extension] Bacterial fermentation can also cause gas and bloating.

Enterokinase = enteropeptidase: it activates trypsinogen rather than serving as a bulk protein-digesting enzyme. Protein transporter mechanisms are outside this lecture's stated absorption outcomes.

13 / Sugar absorption: membrane and energy

Lecture PDF p. 22 | Case 2 Part 4, Q1–2



Original enterocyte transport diagram with epithelial-cell annotation (p. 22; Berne & Levy, 8e, Fig. 30.9).

Membrane / transporter	What moves?	Energy / direction
Apical SGLT1	Glucose or galactose together with Na^+, from lumen into enterocyte.	Secondary active transport: uses the Na^+ electrochemical gradient.
Apical GLUT5	Fructose from lumen into enterocyte.	Facilitated diffusion; Na^+ independent.
Basolateral GLUT2	Glucose, galactose and fructose leave toward blood.	Facilitated diffusion.
Basolateral Na^+/K^+-ATPase	Na^+ out of cell, K^+ into cell.	Uses ATP directly; keeps intracellular Na^+ low to support SGLT1.

Trace the source of energy

ATP → Na^+/K^+-ATPase → low intracellular Na^+
→ Na^+ entry through SGLT1 pulls glucose / galactose into the cell.
SGLT1 itself does not hydrolyse ATP. GLUT5 does not use Na^+ cotransport.

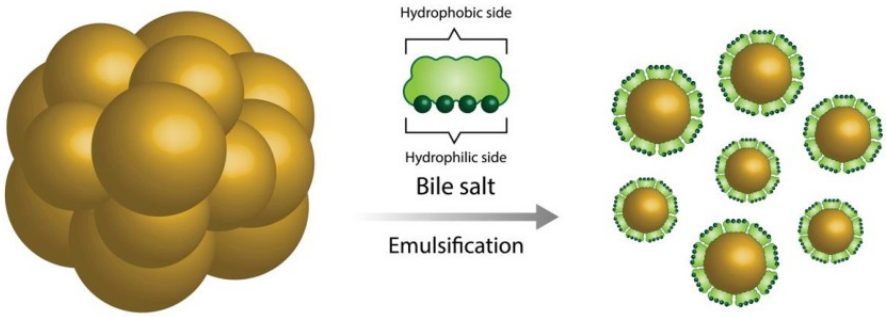
Original Case 2 | Part 4: Nutrient Absorption

Original question / ID	Answer and reasoning
C2-P4-Q1 In what form are carbohydrates absorbed?	Monosaccharides: glucose, galactose and fructose. Starch and disaccharides need prior digestion.
C2-P4-Q2 Which transporter absorbs glucose from the intestinal lumen?	Apical SGLT1. GLUT2 is the basolateral exit transporter, so membrane location determines the answer.

SGLT1 = sodium–glucose cotransporter 1; GLUT = glucose transporter family. Apical faces the lumen; basolateral faces interstitium / blood. The slide's "fructose relatively inefficient" is not a universal quantitative rule: focus on Na^+-independent diffusion versus Na^+-coupled uptake.

14 / Lipid digestion: emulsification is not hydrolysis

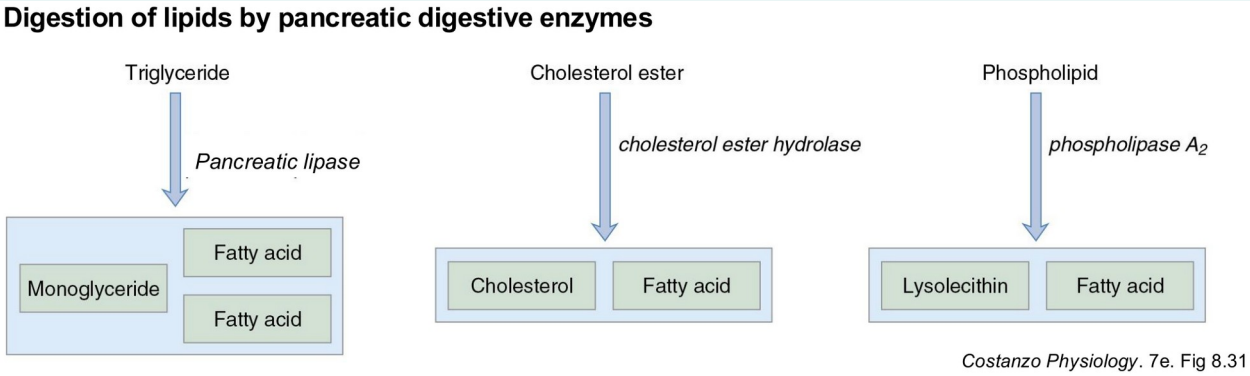
Lecture PDF p. 23 | Case 2 Part 4, Q3



Original bile-salt emulsification diagram (p. 23).

Bile and enzymes have different jobs

Liver makes bile → gallbladder stores and releases bile.
Bile salts emulsify large fat droplets → smaller droplets → surface area increases.
Pancreatic enzymes hydrolyse lipid bonds at the droplet surface.
Bile salts are not digestive enzymes.



Original pancreatic lipid-enzyme pathways (p. 23; Costanzo Physiology, 7e, Fig. 8.31).

Dietary substrate	Enzyme / helper	Main products
Triglyceride (TG)	Pancreatic lipase; colipase helps lipase act at the lipid interface.	2-monoglyceride (2-MG) + two free fatty acids (FFAs).
Cholesteryl ester	Cholesterol ester hydrolase.	Free cholesterol + FFA.
Phospholipid	Phospholipase A_2.	Lysophospholipid, e.g. lysolecithin + FFA.

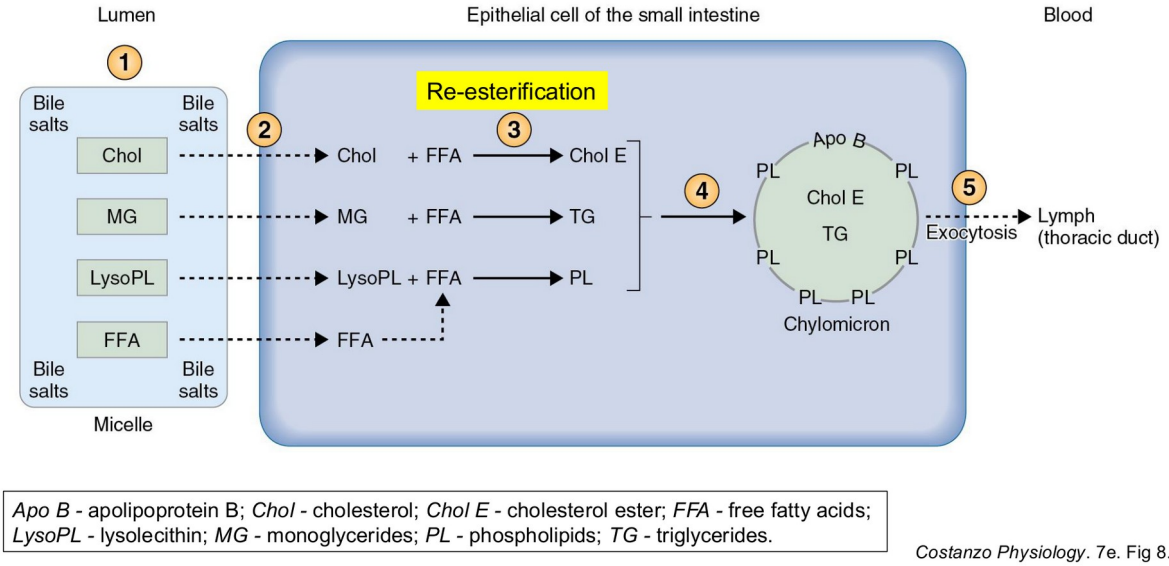
Original Case 2 | Part 4, Q3

Original question / ID	Answer and reasoning
C2-P4-Q3 Into what products are dietary lipids digested?	Main expected answer: FFAs and monoglycerides. For all lipid classes, also include free cholesterol and lysophospholipids.

Digestion makes small lipid products; absorption then requires their delivery to the enterocyte surface. Mixed micelles solve that next luminal transport problem (p. 15).

15 / Lipid absorption: the complete route

Lecture PDF pp. 24–25 | Original MCQ 7; Case 2 Part 4, Q4



Apo B - apolipoprotein B; *Chol* - cholesterol; *Chol E* - cholesterol ester; *FFA* - free fatty acids; *LysoPL* - lysolecithin; *MG* - monoglycerides; *PL* - phospholipids; *TG* - triglycerides.

Costanzo Physiology. 7e. Fig 8.32

Original lipid absorption and re-esterification diagram, with its abbreviations retained (p. 24; Costanzo Physiology, 7e, Fig. 8.32).

Step / location	What happens?
1. Intestinal lumen	Bile salts form mixed micelles carrying FFAs, monoglycerides, cholesterol and lysophospholipids.
2. Apical membrane	Lipid cargo leaves the micelle and enters the enterocyte. Whole micelles do not enter; bile salts remain in the lumen.
3. Inside enterocyte	Lipid products are re-esterified → TGs, cholesteryl esters and phospholipids.
4. Inside enterocyte	Re-formed lipids + apoproteins, including Apo B → chylomicrons.
5. Exit route	Exocytosis → lacteals (intestinal lymphatic vessels) → lymph → thoracic duct → systemic blood.

Original MCQ 7

Dietary lipids are eventually transported in the circulation mainly as:

A. Micelles B. Chylomicrons C. Monoglycerides D. Fatty acids

Answer: B

Chylomicrons carry absorbed long-chain dietary lipids after enterocyte processing. Micelles act in the intestinal lumen, before absorption.

Original Case 2 | Part 4, Q4

Original question / ID	Answer and reasoning
C2-P4-Q4 What structure transports absorbed lipids into the circulation?	Particle: chylomicron. Route: lacteals → lymph → thoracic duct → blood. Give both if “structure” is ambiguous; do not answer “micelle”.

Apo B = apolipoprotein B; Chol = cholesterol; Chol E = cholesteryl ester; PL = phospholipid. “Re-esterification” means joining digestion products back into complex lipids inside the cell; it is not reversal of absorption.

16 / Keep the locations straight: final integration

Lecture synthesis | One further integrated practice question; supporting exceptions are marked [Extension]

Compare	Micelle	Chylomicron
Where formed?	Intestinal lumen.	Inside enterocyte.
Main components	Bile salts + lipid digestion products.	Re-esterified lipids + apoproteins.
Main task	Deliver poorly water-soluble lipids to the apical surface.	Export and carry absorbed dietary lipids.
Does it enter the cell intact?	No; lipid cargo leaves first.	Formed in the cell, then exocytosed.
Next destination	Bile salts stay in lumen, then are largely reabsorbed in ileum.	Lacteals → lymph → thoracic duct → blood.

Bile salts are recycled; chylomicrons take a different route

Bile salts → terminal ileal reabsorption → return to liver → secretion into bile again.

[Extension] This enterohepatic circulation uses portal blood. Long-chain dietary lipids initially leave the intestine in chylomicrons through lymph. Short- and medium-chain fatty acids can enter portal blood more directly.

Location of low pH	Cell / mediator	Response
Stomach	D cells → somatostatin.	Gastric acid secretion decreases: feedback inhibition.
Duodenum	S cells → secretin.	Pancreatic bicarbonate-rich secretion increases; gastric delivery slows.

Integrated practice 2 | Self-written; not an original exam question

No bile reaches the duodenum, but pancreatic enzyme secretion, bicarbonate secretion and the intestinal mucosa remain normal. Explain the effects on long-chain fat absorption, chylomicron output and glucose absorption. Would simply increasing lipase fully fix the problem?

Answer approach:

- Emulsification and mixed-micelle formation are impaired → lipid hydrolysis / delivery to the brush border becomes less effective → long-chain fat absorption and chylomicron output fall.
- Lipase cannot replace bile salts' luminal functions; more enzyme does not fully correct the defect.
- Glucose absorption is largely preserved: starch hydrolysis and SGLT1 / GLUT2 transport do not require bile micelles.
- The question explicitly preserves pancreatic outputs; do not assume every digestive pathway has failed.

Use the tutorial's reasoning pattern

Stimulus → sensor / source cell → neural or hormonal messenger → target → function.

For absorption: final digestion product → apical entry → intracellular processing → exit route.

Source check: the selected MCQs are consistent (B, C, D, C, A, C, B). Case 1 annotations are broadly consistent; use "motility". Buffet "low pH" is corrected on p. 6; its stimulator list is completed on p. 7. Intestinal enzyme-source wording is corrected on p. 12. Figures / highlights are source material, not an official key. The two self-written questions do not predict actual exam content. Main reference named by the lecture: Costanzo Physiology, 7e, Chapter 8; figure sources are credited beside each image.